

U:fynd Assignment by Sandip Hapani

Problem Statement:

Problem was given by the U:fynd – Assignment for the job interview.

During my freelancing days, I always used to be asked my client what's more important built scrappers or Data. The reason behind is if data is more important then some kind of changes in code, we can avoid mostly those are the things which makes scrapper long durable. Otherwise, Scrappers need a high maintenance whenever some changes happened on the targeted site.

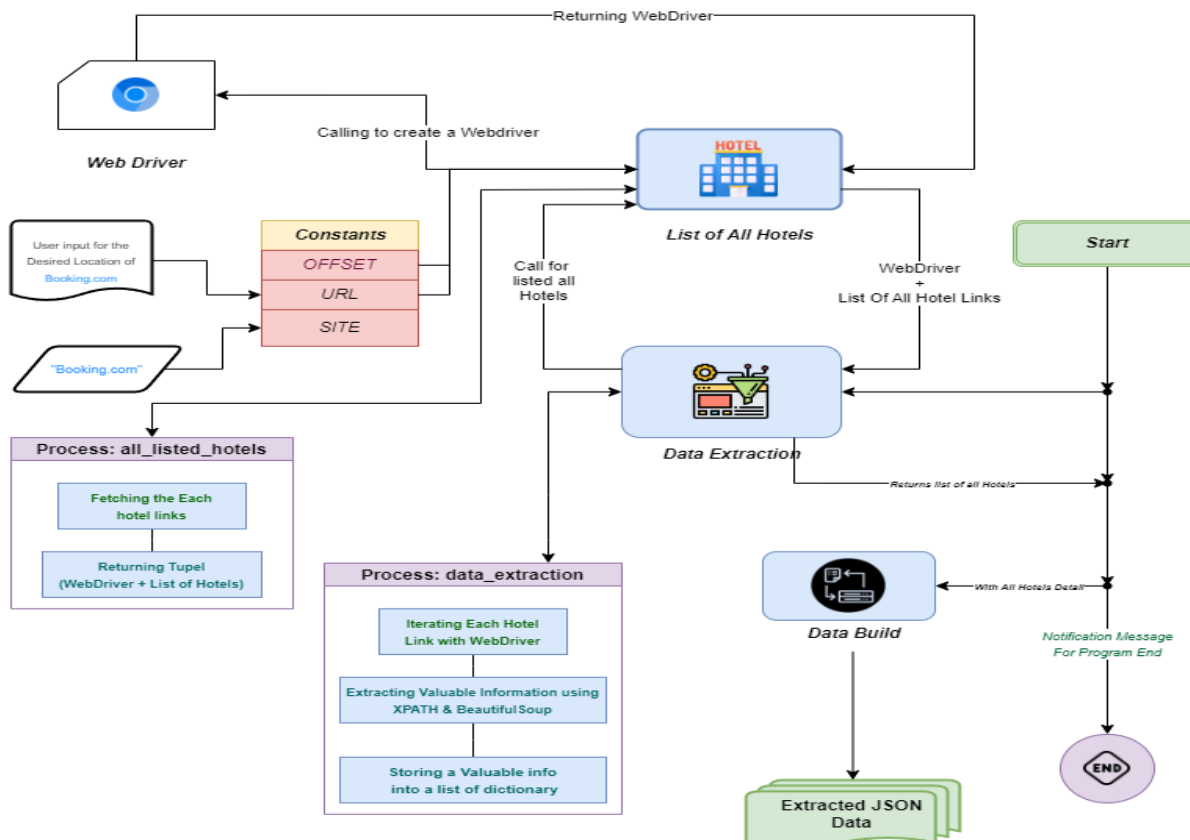
Here, I consider Data as an important part. Obviously, I tried to be built this scrapper dynamic as much as possible.

(Note: I have prepared 2 scrapers for the given task. One was a static scraper which only focus and based on the given web webpage of booking.com site. However, When I checked the site, I noticed some changes which may updated on booking.com over the time. So, I have also prepared the live booking.com site scraping demo.)

Used Tech Stack:

- Domain: Web Scraping
- Programming Language: Python
- Python Libraries: Selenium (For the live Scraper), BeautifulSoup, lxml
- IDE: VS Code

High Level Diagram:



Working Demo:

Static Web Scrapper: https://drive.google.com/file/d/1byz5ibMk-5pvEHA96bQph6_x2z-uP9K/view?usp=sharing

Dynamic Live Web Scrapper: <https://drive.google.com/file/d/1p8qu6-lkGFZrpJgX-qAm4Dmmm75ligkr/view?usp=sharing>

Challenges & How I solved:

I haven't faced any difficulties in Static Web Scrapper – Which was a given task.

Live Web Scrapper:

- Absolute Path selection:
 - This site is bit different from other sites, because here unique identification of HTML tag is not normal. Usually, **class** or **id** is enough to fetch a unique reference of HTML tag.
 - Solution: I found this amazing [website](#) on internet. From which I have deeply sharpen my XPATH concepts.
- Syntactical & Dependencies Errors:
 - Solution: To brush up my syntactical mistakes, I got help from StackOverflow.
- Other than that, I refer Selenium and BeautifulSoup standard documents.
- For the approach Diagram drawing, I took help from [draw.io](#) site.

Future Possible Updates:

- I have tried to make it as much as faster in limited time, but I think still it can be fastened up using **threading**. Here, I have used selenium just for the page response loading. To extract the information, I am using BeautifulSoup which is just parse the response using every HTML tag reference.
Yes, I could have used **request** also which is faster option than selenium but, in that case, we won't get JavaScript rendered web page.
- Here, I can update into a Fetch & Write method. Currently, I am collecting all the data and then I am writing into a file. I can be dangerous because due to any reasons program shutoff then we must loss fetched data also.
- Code Optimization can be possible.
- Here, I haven't covered particular each fields exception, so that can be cover up after complete analysis of every field's behavior at each entry.
- Extension of this scrapper for the whole site's other features also.

Thanks & Regards,

Sandip Hapani